The Power of Stories: Narratives and Information Framing Effects in Science Communication

Yang Yang, and Jill E. Hobbs

This article explores information framing effects by comparing the effectiveness of using logical-scientific versus narrative information to communicate with consumers about a new biotechnology application (gene editing). Using data from an online survey of 804 Canadian adults, a discrete choice experiment elicits preferences for diverse novel food attributes and technologies, with respondents randomly assigned to different information conditions. We construct a logical-scientific information condition, written in a scientific style using the passive voice with generalized and impersonal language and attributed to either a government agency or a scientific organization. In contrast, we frame the narrative-style information condition as a story, using a lively and vivid personal style, and attributed to either a science journalist or a consumer blogger. Data are analyzed using multinomial logit and random parameters logit models. We find that the information format (logical-scientific vs. narrative) matters: narratives help reduce negative perceptions regarding agricultural and food technologies. We also examine factors that predispose consumers to seek logical-scientific versus narrative information sources.

Key words: Biotechnology, discrete choice experiment, gene editing (CRISPR-Cas9), multinomial logit model, narratives, random parameters logit.

JEL codes: D12, D83, D91, Q13, Q18.

Over the past few years, the scientific community, agrifood industry stakeholders, and regulators have made efforts to promote biotechnology communication; however, attitudes toward agricultural biotechnology among the general public have become more polarized and persistent. Controversy around genetically modified (GM) food remains rife despite a flood of information, with deepening division among consumers regarding its safety and desirability (Pew Research Center 2016). Studies have challenged the hypothesis that "more" information facilitates technology consensus and acceptance (Kahan *et al.* 2009), and consistently demonstrate that framing matters for scientific communication (Fox, Hayes, and Shogren 2002; Rousu *et al.* 2007). Journalists, environmental activists, and politicians seem more successful than do scientists in formulating their biotechnology messages to connect with diverse audiences (Nisbet *et al.* 2002; McCluskey and Swinnen 2004), and a distinguishing feature of their messages is the use of narratives and storytelling, which is not typically a medium used by scientists or regulators (Dahlstrom 2014).

Framing is unavoidable in science communication and involves simplifying complex issues by lending greater emphasis to certain aspects over others (Nisbet and Mooney 2007; Bubela *et al.* 2009). Individuals are often characterized as "cognitive misers," rarely well enough motivated to understand a complex science debate, such that only appropriately framed science messages exert desirable influences on attitudes and behaviors (Kahneman 2011). This article

Yang Yang is a faculty lecturer at the Johnson Shoyama Graduate School of Public Policy, University of Saskatchewan. Jill E. Hobbs is a professor at the Department of Agricultural and Resource Economics, University of Saskatchewan. The authors gratefully acknowledge research funding from the Alliance for Food and Bioproducts Innovation Scholars Program (AFBI), Saskatchewan Ministry of Agriculture. The authors would like to thank editor Timothy Richards and three anonymous reviewers for their helpful comments and suggestions.
Correspondence may be sent to: yang.yang@usask.ca.

Amer. J. Agr. Econ. 00(00): 1–26; doi:10.1111/ajae.12078
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contributes to the literature on information framing effects by examining whether narratives (stories) about agricultural biotechnology (specifically gene editing) result in consumer attitudes and choice behaviors that differ from when information is positioned in the logical-scientific frame more commonly used by scientists, regulators, and experts.

Numerous studies have analyzed information framing effects in biotechnology communication. The type of information (pro-biotech vs. anti-biotech), the sequence of information presentation, and the credibility of information sources are all decisive factors shaping consumers' attitudes and willingness-to-pay values for agrifoods produced with biotechnology (Huffman *et al.* 2003; Huffman *et al.* 2004; Lusk *et al.* 2004; Huffman *et al.* 2007; Rousu *et al.* 2007; Rousu and Lusk 2009; McFadden and Lusk 2015). To the authors' knowledge, however, none of these studies has evaluated the power of using narratives and storytelling to communicate biotechnology with non-expert consumers. It is not uncommon in survey and experimental studies that individuals are furnished with exogenous "new" information treatments about biotechnology; however, most, if not all, of the information in previous studies is presented in a logical-scientific format.

A logical-scientific frame typically emphasizes rigorous science research and dissemination of unbiased scientific facts; however, it underperforms when communicating science with the less informed lay public. Narrative formats of communication (stories) are inherently easier to comprehend; more interesting, engaging, persuasive; and are commonly used in media content where consumers get most of their science information (Dahlstrom 2014). Researchers have documented the benefits of narrative communication in diverse non-food contexts, such as communicating the risks of indoor radon (Golding, Krimsky, and Plough 1992), vaccination (Betsch *et al.* 2011), and promoting cancer prevention behaviors (Kreuter *et al.* 2007).

This article is the first to shed light on the effects of using narratives (stories) to communicate about food biotechnology with non-expert consumers. Using an online survey instrument situated in a food purchase decision-making context, we develop a logical-scientific frame describing a gene editing application written in a passive voice, using a generalized and impersonal tone. In contrast, we also develop a narrative frame featuring a more lively and vivid personal style. We compare the influence of the two communication approaches on consumer food choice behaviors through a discrete choice experiment. This article examines why consumers respond differently to different information formats and sources (including being assigned to or self-selecting different types of information), which has received scant attention in previous literature.

The technological focus of the study is gene editing (also known as CRISPR-Cas9), as such, the article also provides a timely consideration of consumer responses to gene editing, which can be applied in plant and animal agriculture, as well as human health. The adoption of gene editing in agriculture and food production has triggered divergent policy and regulatory responses.[i] Unlike genetic modification, which often requires inserting foreign genes from other species, gene editing increases the ease, precision, and speed of genetic improvements by precisely targeting and editing genes, such as switching on or off a specific trait (Ledford 2015). However, public opinions of gene editing remain nascent, and there is limited information about how consumers will respond to the use of this new biotechnology tool in agriculture and food production.

In addition to a timely evaluation of gene editing technology, this article offers a more nuanced understanding of information framing effects in public biotechnology communication. The power of narrative communication is often neglected in an agricultural or food context. We find that, despite a reluctance to accept novel food technologies, consumers soften their initial negative reactions to gene editing faced with information about the technology framed in a narrative format or accessed voluntarily. This article also sheds light on why consumers choose to access a particular format or source of information. Our results show that individuals holding more favorable views toward science, and who are more familiar with biotechnology, are predisposed to seek the logical-scientific (vs. narrative) information source from a government agency or a scientific body, whereas socio-demographic characteristics and trust have little impact on information choice.

The next section synthesizes studies on information framing effects in biotechnology communication and the use of narratives in non-food contexts. Consumer data are

[i] In July 2018, the European Court of Justice ruled that gene editing was a form of genetic engineering and therefore subject to the same set of regulations as genetically modified crops, but in the U.S. the new federal labeling requirement for genetically modified (bioengineered) foods does not cover gene edited crops.

collected from an extensive online survey, administered in Canada, and featuring information treatments (logical scientific, narrative, and self-selection) and a choice experiment (prepackaged apple slices with consumer-oriented food traits [non-browning and antioxidant enhanced] introduced through food technologies [gene editing, genetic modification, and edible coating]). The Experimental Design and Data Collection section details the survey sample characteristics and the experimental design, including a contrast between the narrative and logical-scientific information treatments. The Econometric Models section specifies the multinomial logit and random parameters logit models used for estimation. The Results and Discussion section compares perceptions of information framed in a narrative versus a logical-scientific rationale, disentangles the factors influencing why respondents choose different information sources, and explores narrative effects in food choice behaviors. In light of the analysis, the final section concludes and explores implications for stakeholders concerned with effective public communication of agricultural biotechnology.

Insights from the Literature

Despite being reassured about their safety, many consumers remain wary of foods produced using biotechnology (Lusk *et al.* 2005; Costa-Font, Gil, and Traill 2008). A recent study suggests that 37% of the U.S. public believed genetically modified foods are safe to eat (Pew Research Center 2015). This figure remains similar among Canadians (37.7%, Charlebois *et al.* 2019); however, it drops dramatically for Europeans (23%, Gaskell *et al.* 2010). Consumers possess a relatively low level of knowledge and understanding of the technology (McFadden and Lusk 2015), leading to the "knowledge deficit" hypothesis that *more* information or education will develop consensus between the public and the majority of scientists who defend the safety of agricultural biotechnology (Allum *et al.* 2008). In a nanotechnology context, provision of more information, however, failed to achieve a broader level of agreement among the public, instead a greater polarization in attitudes emerged (Kahan *et al.* 2009).

Previous research has shown that a plethora of factors influence individuals' processing and assimilation of provided information, such as prior beliefs about the technology, underlying worldviews, and values. For example, Huffman *et al.* (2007) detailed the impacts of prior beliefs on consumers' acceptance of genetically modified foods. Individuals with well-established prior beliefs were more immune to "new" information from interested or independent third parties than uninformed individuals. McFadden and Lusk, McFadden, and Rickard (2015) suggest that individuals only assimilate new "scientific" information about genetic modification food when it conforms to their prior beliefs. Kahan *et al.* (2009) demonstrated that, with exposure to the same balanced information about nanotechnology, attitudes become more divergent among people with competing cultural worldviews.

Additionally, in novel food technology communication, studies have shown that *how* the communicator presents the information is as important as the content of that information (i.e. information framing effects). For instance, negatively framed information about food irradiation dominates favorably framed information in reducing consumers' willingness to pay for an irradiated pork sandwich (Fox, Hayes, and Shogren 2002). Huffman *et al.* (2003) and Rousu *et al.* (2007) identify a similar effect with genetically modified foods; however, the dominant impact of negative information from interested parties became muted with the addition of verifiable third-party information. The sequence of information also plays an important role in attitude formation (Marette *et al.* 2008; Roosen *et al.* 2011). Lusk *et al.* (2004) showed that different types of positive biotechnology information – benefits to environment, health, and developing countries – affect consumers' valuations of a GM cookie to different extents. Subsequent work by Rousu and Lusk (2009) revealed that environmental information is most effective at switching consumers from purchasing non-GM to GM cookies; however, switchers place greater value on information pertaining to health and benefits to developing countries.

Furthermore, *who* provides the biotechnology information also matters for its acceptance. Consumers often perceive independent third parties and government as more reliable information sources compared to interested groups (e.g. industry or environmentalists; Frewer, Howard, and Shepherd 1998; Huffman *et al.* 2004). However, information from the most trustworthy sources seems to only reach a limited audience, as most consumers learn about food science and technologies from diverse media outlets, including television, radios,

newspapers, and other Internet sources such as social media and online blogs (Agriculture and Agri-Food Canada 2014; National Science Board 2016).

Media coverage of biotechnology tends to be dominated by negative frames, which has been criticized as one of the driving forces behind public resistance to agricultural biotechnology (Kalaitzandonakes, Marks, and Vickner 2004; McCluskey and Swinnen 2004; McCluskey, Kalaitzandonakes, and Swinnen 2016), with individuals inherently overemphasizing small probability negative events (Kahneman and Tversky 1979). Concerns about the potential risks of biotechnology become readily available in consumers' memories (known as the "availability heuristic") with repeated sensationalized media stories (Lusk, Roosen, and Bieberstein 2014). Although accuracy and scientific rigor are critical in scientific research, the principal practice for developing science news is to personify abstract scientific concepts for dramatic storytelling, often at the expense of accuracy (Dahlstrom 2014). Indeed, using emotional narratives (stories) is one of the major characteristics of media content, which helps journalists attract more attention from their target audience (Dahlstrom 2014).

Individuals are boundedly rational, with limited cognitive capacities and/or motivations to process complex scientific information (Simon 1955; Tversky and Kahneman 1974). As such, information framing and presentation matters in shaping attitudes and behaviors (Tversky and Kahneman 1981). Although prior research has examined information framing effects in agricultural biotechnology communication, exploration of narrative communication effects is limited to non-agricultural contexts.

Narrative communication can take various forms, such as news stories, anecdotes, entertainment television programs, and interpersonal conversations. A fairly standard definition of narrative communication is given by Dahlstrom (2014, p. 13614): "a particular structure that describes the cause-and-effect relationships between events that take place over a particular time period that impact particular characters." As such, three essential factors – causality, temporality, and character – determine the narrativity of a message.

Narratives (stories) stand in contrast to logical-scientific communication, which is usually abstract, expository, argumentative, statistical, and technical (Norris *et al.* 2005; Dahlstrom 2014). Unlike logical-scientific messages, which define propositions or explain a list of facts or abstract concepts, narratives aim to explain a series of causally linked events that unfold over time (Norris *et al.* 2005). People are more familiar with the narrative format of communication, as it represents the default mode of human interaction and characterizes most daily conversations. In comparison, logical-scientific communication usually appears "cold," using vocabulary and an expository writing structure with which many people have less experience. People may thus feel more comfortable with the use of narratives (stories) to communicate complex scientific issues.

Messages framed in a logical-scientific or narrative format are likely to provoke and engage distinct systems of cognitive reasoning possessed by individuals (Kahneman 2011). Encoding science-based arguments requires the deliberative, logical, and slow cognitive pathway, whereas encoding situation-based narratives requires the intuitive, affective, and fast pathway (Dahlstrom 2014). Compared with logical-scientific information, narratives are found to offer shorter reading time; increased levels of comprehension, engagement, persuasiveness, and recall of content over time (Avraamidou and Osborne 2009; Dahlstrom 2014).

Narrative communication is powerful in non-food contexts such as science education and health communication. For example, Golding, Krimsky, and Plough (1992) revealed that narratives are more effective than logical-scientific information to communicate with the general public about the environmental risk of radon. Betsch *et al.* (2011) found that, in an online bulletin board setting, stories have a greater influence than statistical information on individuals' perceived risk of side effects from vaccination and thus their intention to vaccinate. Evidence also exists for narrative effects in promoting healthier behaviors and communicating health-related issues that often encounter resistance, such as cancer and human papillomavirus (HPV) vaccination (Kreuter *et al.* 2007; Mazor *et al.* 2007; Nan *et al.* 2015).

Despite the merits of narrative communication and its documented effectiveness in teaching science and in communicating health issues, studies on the acceptance of food technologies have overlooked this information format, in most cases assigning participants to information treatments written using a logical-scientific rationale. This article fills this gap in the literature by investigating whether a narrative effect exists when communicating a controversial agricultural biotechnology with non-expert consumers. The narrative developed in this article entails a mother (i.e. the narrator)

depicting how her family encountered a new apple variety produced using novel food technologies and their reactions to this apple. The next section details the design of the online consumer survey used to gather data for the study and discusses the development of the logical-scientific and narrative information used in the information treatments.

Experimental Design and Data Collection

This section details the survey sample characteristics and the experimental design, including the information treatments and a choice experiment.

Consumer Data

About 800 Canadian adults completed the online survey between July and September 2016, with 19.5% completing in French and the remaining in English.[ii] To ensure necessary experience with the food purchase featured in the choice experiment, only primary shoppers in a household participated in this study. The online survey consists of two key sections – information treatments and a choice experiment featuring prepackaged apple slices. Each participant made six choice decisions in the choice experiment, resulting in a final sample of 804 respondents and 4,824 choice observations. The sample is reasonably representative of the Canadian population by gender and province, however, leans toward a slightly older, better educated, and higher income demographic, which is common in web-based surveys (Szolnoki and Hoffmann 2013). Table 1 summarizes the sample characteristics.

Survey Design

The central aim of this article is to explore whether consumers respond differently toward new information about biotechnology framed in a logical-scientific versus narrative style. The online survey randomly allocated respondents into one of four information conditions[iii]: (*a*) "no additional information (control)", where respondents were exposed to only a generic description about biotechnology but no further detailed information; (*b*) a "logical-scientific" condition, where respondents received additional "logical-scientific" style information about the food technologies featured in the subsequent choice experiment; (*c*) a "narrative" condition, where equivalent information was framed instead in a "narrative" style was provided to respondents; (*d*) a "self-selection" condition, where respondents were permitted to choose one piece of information from four available sources: Health Canada, The Royal Society of Canada, a consumer blog post, and a science journalist's article published in *The Globe and Mail*, with the first two sources framed in a "logical-scientific" style and the latter two as "narratives." Figure 1 illustrates the number of respondents assigned to each information condition.

Respondents were provided with information on food biotechnology explaining how gene editing differs from genetic modification. The "logical-scientific" information condition contained factual scientific content synthesized from resources such as scientific reports and announcements from scientists and governmental authorities (e.g. Health Canada and the Canadian Food Inspection Agency). The "logical-scientific" information deliberately utilized scientific language and a plain tone, purposely avoiding any metaphorical descriptions. Both scientists and regulatory agencies represent the primary sources disseminating "logical-scientific" information, thus the same information was attributed to two sources in different conditions: "Health Canada," the regulating body for biotechnology applied to foods in Canada; and "Academy of Science | The Royal Society of Canada," a community of distinguished scientists.

The "narrative" information condition deliberately featured less technical language with a vivid tone and was adapted from a blog article written by Emily Waltz, a science journalist specializing in writing science stories

[ii]The study received ethics approval (BEH# 16–131) from the University of Saskatchewan Behavioral Research Ethics Board in May 2016. The Social Sciences Research Laboratories (SSRL) at the University of Saskatchewan administered the online survey. Respondents were recruited through a third party research vendor, EKOS, who recruits its panel members exclusively through a random digit dialing process from a blended land-line cell-phone frame. EKOS's member panel is meant to mirror the actual Canadian population as defined by Statistics Canada. Our sample is regionally representative as it was selected from among panel members using quotas on province distribution. As a means to incentivize participation, respondents entered into a draw to win one of two prizes of $500 upon completion. The full survey instrument is available in Yang (2018).

[iii]See Appendix A for full scripts of all information conditions. In anticipation of respondents being unfamiliar with biotechnology (65% of sampled respondents), a neutral, generic introductory information on agricultural biotechnology was furnished to all respondents regardless of information conditions.

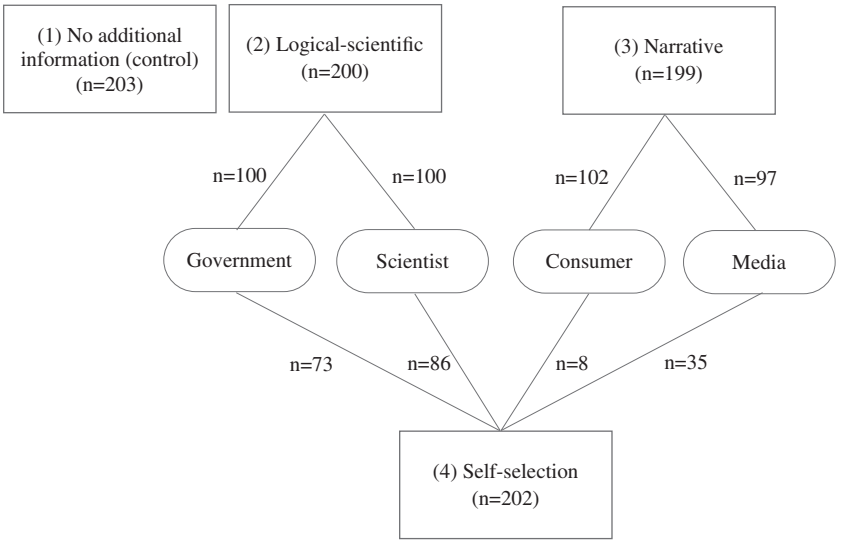

Figure 1. Allocation of respondents to information conditions, n = 804

Table 1. Sample Characteristics

Variable	Sample (n = 804)	Canadian population[1]
Gender (%)		
Female	52.9	50.9
Male	47.1	49.1
Age (mean)	54.5	47.2[2]
Highest education completed (%)		
Below Bachelor's degree	46.6	71.5
Bachelor's degree and above	53.4	28.5
Annual household income, before tax (%)[3]		
$49,999 and under	24.1	34.8
$50,000 - $124,999	52.5	43.2
$125,000 and over	23.4	22.0
Children in household (%)		
Yes	24.8	–
No	75.2	–

[1] Data from Canada Census 2016 (Statistics Canada).
[2] Mean for Canadians aged 15 years and above.
[3] Income indicated in Canadian dollars (CAD). Exchange rate: 1 CAD = 0.76 USD / 0.68 EUR (July 2019).

(Waltz 2015). The story-telling context and wording adopted in the narrative format resemble those commonly seen in blog posts and news stories. Hence, the survey attributes the same narrative information to two sources in different information conditions: a regular consumer who reports her experience with novel food technologies in a blog post and a science journalist who writes for a national newspaper *The Globe and Mail*. Respondents assigned to the "logical-scientific" or "narrative" condition were further randomly allocated to one of two information sources. Table 2 summarizes four key features that distinguish the "logical-scientific" and "narrative" information treatments developed in this study.

The "logical-scientific" and "narrative" messages adopt different writing styles. The "logical-scientific" frame aims to inform respondents through a list of abstract facts about biotechnology; hence, it appears to be expository, impersonal, and uses technical vocabulary and passive voice, consistent with the communication style typically used by regulatory agencies or expert scientists. By contrast, the "narrative" frame aims to explain

Table 2. Distinguishing Features of "Logical-Scientific" versus "Narrative" Information Treatments

	Logical scientific	Narrative
Genre	Expository to explain abstract scientific concepts	Tells a story about a family's encounter with food technology
Source	Scientific reports, governmental announcements	Consumer blog posts, news stories by journalists
Language	Scientific, impersonal, technical words	Vivid, metaphors
Tone	Passive, plain	First-person, active

Table 3. Attributes and Levels in Choice Experiment

Attribute	Levels			
Appearance	Non-browning	Slices turn brown		
Health Benefit	Enhanced with antioxidants like Vitamin C	Not enhanced with antioxidants		
Production method	Gene editing	Genetic modification	Edible coating	Conventional
Price	$3.69	$4.29	$4.89	

biotechnology to respondents through a series of events in an attractive storytelling context. Specifically, the first person narrator in the "narrative" tells a story of home experiments with an apple produced with the use of biotechnology. Thus, the "narrative" message uses vivid language and a first-person, active voice. To illustrate differences in the two information treatments: the "logical-scientific" information describes gene editing as "*it locates a gene to be edited, then makes the necessary changes, either a deletion or a repair*," whereas the "narrative" information employs a metaphorical description "*gene editing technology, which locates the gene to be edited and then makes necessary changes, resembling the find-and-replace function on the computer*." Further, to introduce the new biotech apple varieties, the "logical-scientific" information expresses the information in a "cold" manner: "*agricultural biotechnology is a collection of scientific techniques used to improve crops and foodstuffs. Scientists have been using biotech tools to develop new apple varieties with desired characteristics, such as non-browning apples and apples enriched with antioxidants*"; whereas the "narrative" starts with an engaging story prologue: "*innovations in biotech crops aren't known for getting families excited, but there are new fruit traits that families may actually notice and appreciate. They are apples that don't turn brown when sliced and are enriched with healthy antioxidants. My children and I had a chance to try some last week and see how child friendly and tasty they really are — a snack and a science lesson in one*."

A manipulation check was conducted immediately after the information treatments. We evaluated the effects of information format and the manner of accessing information (assigned vs. self-selection) on respondents' reading time and the perceived quality of the information. The latter included "easiness to understand," "persuasiveness", "trustworthiness", "source credibility," and "factualness." We detail results of the manipulation check in a subsequent section. Despite differences in writing style, efforts were also made to maintain content consistency across the "logical-scientific" and "narrative" formats; that is, both messages describe novel food technologies and explain how they differ in achieving desirable food characteristics. Both messages display in a comparable length and layout, and were pretested by a small pilot group to check the plausibility of language, situations, and characters employed in both messages.

Following the information provision, respondents made a series of food choices in a choice experiment featuring a set of 500 g bags of prepackaged apple slices varying in four attributes: (*a*) *appearance* of apple slices, apples turn brown in minutes after being sliced

Table 4. An Example of a Choice Set

Imagine that you are actually buying a 500g bag of apple slices in a real grocery store. If you were able to select from the following options, which one would you buy?

	A	B	C	D
Appearance	Non-browning	Slices turn brown	Slices turn brown	I would not buy any of these products
Health benefit	Not enhanced with antioxidants	Enhanced with antioxidants like Vitamin C	Not enhanced with antioxidants	
Production method	Genetic modification	Gene editing	Conventional	
Price	$4.29	$4.89	$3.69	
I would choose…	☐	☐	☐	☐

or resist browning for a long time; (*b*) *health benefit*, sliced apples are enhanced with higher levels of dietary antioxidants like Vitamin C or contain only a regular amount of antioxidants; (*c*) *production method*, the aforementioned two new apple characteristics, non-browning and antioxidant enhanced, are introduced by three different food technologies – gene editing, genetic modification, or edible coating. Apples produced via conventional methods will not possess either of the two new characteristics; and (*d*) the retail *price* for a 500 g bag of apple slices. Table 3 summarizes the apple attributes and levels included in choice experiment, and table 4 presents an example of choice set.

Consumers appear to be more accepting of biotechnology when direct and tangible benefits are provided (Hossain *et al.* 2003; Colson and Huffman 2011; Lusk, McFadden, and Rickard 2015; Lusk, McFadden, and Wilson 2018). The recent regulatory approval of the Arctic Apple in Canada and the U.S, which is one of the first genetically modified foods marketed directly to consumers, inspired the inclusion of the non-browning feature.[iv] The inclusion of a health attribute (antioxidant-enhanced) allows us to further explore consumers' tradeoffs between tangible benefits and the use of novel food technologies, as there exists mixed evidence regarding consumers' valuations of fruits and vegetables enriched with antioxidants and Vitamin C (Markosyan, McCluskey, and Wahl 2009; Colson, Huffman, and Rousu 2011). In addition, the nature of the specific biotechnology technique seems to matter for its acceptance, with consumers being more resistant to the transgenic than the intragenic/cisgenic GM method (Colson, Huffman, and Rousu 2011; Delwaide *et al.* 2015). The extent to which consumers differentiate among gene editing, traditional genetic modification, and a food processing technique (edible coating), and the role of information framing (narratives vs. a logical-scientific rationale) in influencing their food technology perceptions were therefore of interest.

We employ a partial constant design, with each choice set consisting of two alternatives with consumer-oriented apple benefits introduced by food technology, one conventional alternative without any additional appearance or health benefits, and one no-purchase option. We retract dominant and implausible alternatives from the design. For instance, we consider alternatives with the antioxidant-enhanced characteristic but lower price levels as dominant if levels of other attributes are constant. In addition, novel production methods (gene editing, genetic modification, and edible coating) have to be associated with at least one of the consumer-oriented apple benefits (non-browning, antioxidant enhanced, or both). The experimental design does not permit the conventional production method to appear with either of the enhanced apple traits. The unlabeled *D-efficient* design was generated using SAS experimental design and choice modeling macros *%ChoicEff* and *%MktBlock* (Kuhfeld 2005). We partition the thirty-six choice situations into six blocks, with each participant being randomly assigned into a block and making six choices.

Econometric Models

We identify information framing effects in choice behavior by comparing preferences

[iv]The Arctic Apple is genetically modified using the RNA-interference technique, which silences the gene expression of enzymes cause apple browning (Okanagan Specialty Fruits 2013).

and willingness-to-pay (WTP) values obtained from econometric models estimated for each information treatment. According to the random utility model (McFadden 1974), the utility that individual n receives from alternative j in choice set s consists of a deterministic component V_{nsj}, and a random component ε_{nsj} that captures all unobserved effects.

$$(1) \quad U_{nsj} = V_{nsj} + \varepsilon_{nsj}$$

The utility-maximizing individual would choose an alternative that yields the highest utility level. The standard multinomial logit (MNL) model assumes the error term to be identically and independently distributed as the extreme value type 1 (EV1) distribution. Following Train (2009), the MNL choice probability is:

$$(2) \quad P_{nsi} = \frac{exp(\lambda V_{nsi})}{\sum_j exp(\lambda V_{nsj})} = \frac{exp(\lambda \beta x'_{nsi})}{\sum_j exp(\lambda \beta x'_{nsj})}$$

where the systematic portion of the utility ($V_{nsj} = \beta x'_{nsj}$) is assumed linear in preference parameters (β) and determined by a vector of apple slices' attributes (x_{nsj}). λ is the scale parameter inversely related to the variance of random component ($\lambda = \sqrt{\frac{\pi^2}{6}}\frac{1}{\sigma_\varepsilon}$) and not identifiable separately from preference parameters in any single data set, thus it is conveniently normalized to one. As such, the observed differences in parameter estimates ($\lambda\beta$) across information treatments may result from differences in the true underlying preference parameters and/or differences in variance (Swait and Louviere 1993).

To evaluate the information framing effects across information treatments and to justify the pooling/combining of data over certain treatments, we utilize the two-step statistical test proposed by Swait and Louviere (1993), who suggested that the *relative* scale parameter(s) can be identified when two or more data sets exist. More specifically, we first separately estimate a MNL model for two information treatments and obtain the likelihood values (L_1, L_2). Then, we estimate the joint model by rescaling the treatment 2 data relative to treatment 1. We utilize a manual grid search approach to estimate the relative scale factor that maximizes the joint likelihood ($\bar{\lambda}$, L_λ) (Swait and Louviere 1993; Adamowicz, Louviere, and Williams 1994; Louviere, Hensher, and Swait 2000, p. 237–240). The null hypothesis of preference and scale equality is rejected if the joint likelihood (L_λ) differs significantly with the sum of separate likelihoods ($L_1 + L_2$).

To relax the IIA assumption and accommodate the panel nature of choice data (each respondent made six consecutive choices), we also estimate a random parameters logit (RPL) model. The RPL model allows for heterogeneous preferences by specifying individual-specific preference parameters:

$$(3) \quad \beta_{nk} = \beta_k + \sigma_k v_{nk}$$

where β_k is the mean preference for attribute k, which stays constant over individuals, and σ_k is the standard deviation of the distribution of β. v_{nk} is the random component of preference, which is assumed standard normally distributed. In this article, we assume the random parameters of all non-price attributes are independently normally distributed, while price parameters are fixed.

As in the case of the MNL model, the RPL preference estimates are compared across information treatments, however, without considering the differences in scale. The estimation of relative scale parameters in the RPL model is problematic because the likelihood function is not globally concave with the relative scale factor (Lusk and Schroeder 2004). To overcome this challenge, we compare the WTP values obtained from the RPL model across information treatments. We calculate the WTP estimates as the ratios of preference parameters for non-price attributes to the price parameter.

$$(4) \quad WTP_k = -\frac{\beta_k}{\beta_{price}}$$

In WTP calculations, the confounded scale parameter is canceled out as it remains constant for any given information treatment. For a given information treatment and apple attribute, we simulate the distribution of 1,000 WTP estimates using the parametric bootstrapping procedure suggested by Krinsky and Robb (1986). We use the complete combinatorial approach proposed by Poe, Giraud, and Loomis (2005) to test the differences in WTP distributions. More specifically, the non-parametric combinatorial test compares all possible differences between the 1,000 bootstrapped WTP values from treatment 1, to the 1,000 bootstrapped WTP values from the treatment 2, resulting in a total of

1,000,000 differences. The proportion of non-positive values is the p-value of the one-sided test that the WTP distributions are equivalent across two treatments. We implement all the econometric models and statistical tests mentioned using Nlogit 6.

Results and Discussion

As illustrated in figure 1, there exist nine information treatments – no additional information, assigned logical-scientific information sourced from the government or a scientific organization, assigned narrative information sourced from either a consumer blogger or a newspaper journalist, self-selected logical-scientific information sourced from a government agency or scientific organization, and self-selected narrative information sourced from a consumer blogger or a journalist. The upper panel of table 5 reports the summary statistics of respondent socio-demographic characteristics across these treatments; however, we identify no systematic differences across information treatments.

Perception of New Biotechnology Information

The survey contains a manipulation check immediately after presentation of the information conditions. Except for the "no additional information (control)" condition, respondents were furnished with, or actively selected, a piece of information about biotechnology presented in a logical-scientific or a narrative format (Appendix). In the "assigned" condition, where respondents read a randomly assigned message, narratives took a slightly longer time to read than information framed in a logical-scientific format (1.66 vs. 1.42 minutes).[v] However, this difference becomes negligible in the "self-selection" condition (1.54 vs. 1.51 minutes).

Immediately after the information treatments, respondents evaluated the quality of the information by indicating to what extent the message was "easy to understand," "persuasive," "trustworthy," "from a credible source," and "factual and unbiased" based on a six-point Likert scale (1 = *not at all*, 6 = *extremely*). Figure 2 visualizes the perceived message quality across treatments.[vi]

In both the "assigned" and "self-selection" conditions, respondents perceived the narrative message as "easier to understand," whereas they considered the logical-scientific message more "trustworthy" and from a more "credible source."[vii] More interestingly, respondents rate the same logical-scientific message as "easier to understand," more "trustworthy," "credible," and "factual" if accessed voluntarily (self-selection) compared to passively (assigned). As such, the factually equivalent information about biotechnology framed in different formats (logical scientific vs. narrative) or accessed by respondents through different pathways (random allocation vs. voluntary choice) can induce differing quality perceptions with respect to the message.

Insights from the Self-Selection Condition

Before exploring the impacts of logical-scientific and narrative biotechnology messages on food choice, we probe the factors influencing the choice of information source in the "self-selection" condition where 202 respondents voluntarily access one piece of information framed with either a logical-scientific or a narrative format (Appendix , Information Condition 4). Table 5 details the differences in individual respondent characteristics among those choosing different information sources: a government agency, scientific body, consumer blogger, or journalist. At a 5% significance level, none of the characteristics of respondents selecting "government" are significantly different with those selecting "scientist." Similarly, no systematic differences exist between respondents selecting "consumer" and "journalist." Further t-tests (5% level) reveal that, compared to respondents who chose the "narratives," those selecting the "logical-scientific" format hold relatively more favorable attitudes toward science and technology in general (*Science*), self-reported as more familiar with agricultural biotechnology (*Familiar*), and had a greater level of trust in government or scientists

[v]The difference in reading time is only significant at a 5% level. We made efforts to maintain a comparable length and layout in both information formats, with the logical-scientific message containing 305 words presented in 5 paragraphs, and the narratives having 370 words shown in 6 paragraphs.

[vi]*T-tests* suggest equivalency across source treatments (See Appendix B, Table B1 for more detail). That is, respondents perceived the logical-scientific information sourced from a government agency or scientific body as similarly "easy to understand," "persuasive," etc. This holds true for the narratives attributed to a consumer blogger or newspaper journalist.

[vii]The Bonferroni procedure (Armstrong 2014) was used to conduct multiple-comparison tests across information treatments at a 5% significance level.

Table 5. Comparison of Individual Characteristics, Prior Beliefs, and Choice Decisions across Information Treatments

			Assigned				Self-selection			
			Logical scientific		Narrative		Logical scientific		Narrative	
Variable	Description	No additional information (n = 203)	Government (n = 100)	Scientist (n = 100)	Consumer (n = 102)	Journalist (n = 97)	Government (n = 73)	Scientist (n = 86)	Consumer (n = 8)	Journalist (n = 35)
Socio-demographic										
Male	1 if male; 0 otherwise	0.51	0.37	0.43	0.52	0.50	0.51	0.47	0.63	0.29
Age	Age in years	53.7	55.3	52.3	54.2	56.2	55.5	55.2	48.0	57.4
University	1 if obtained university degree or higher; 0 otherwise	0.52	0.46	0.54	0.51	0.49	0.62	0.64	0.38	0.59
Inc	1 if annual household income greater than CAD$125,000; 0 otherwise	0.28	0.26	0.21	0.23	0.11	0.26	0.27	0.13	0.21
Kid	1 if children under 18 years old live in household; 0 otherwise	0.23	0.24	0.34	0.22	0.20	0.27	0.28	0.50	0.18
Biotechnology beliefs										
Science	Attitudes to science and technology in general: 1 = the world is lot worse off because of S & T; 6 = the world is lot better off because of S & T						4.95	5.22	4.63	4.57
Familiar	Familiarity with biotechnology:1 = know some or a lot about agricultural biotechnology; 0 = know nothing at all or just a little						0.33	0.45	0.25	0.23
Risk	1 if believe the risks of biotechnology outweigh benefits; 0 otherwise						0.30	0.30	0.50	0.40

(Continues)

Table 5. Continued

Variable	Description	No additional information (n = 203)	Assigned: Logical scientific: Government (n = 100)	Assigned: Logical scientific: Scientist (n = 100)	Assigned: Narrative: Consumer (n = 102)	Assigned: Narrative: Journalist (n = 97)	Self-selection: Logical scientific: Government (n = 73)	Self-selection: Logical scientific: Scientist (n = 86)	Self-selection: Narrative: Consumer (n = 8)	Self-selection: Narrative: Journalist (n = 35)
Trust in biotechnology information sources										
Government	1 = no trust at all; 6 = very high trust						3.88	3.69	2.75	3.23
Scientist							4.16	4.27	3.46	3.96
Consumer							3.81	3.94	3.75	3.97
Science journalist							4.21	4.35	3.50	4.17

	No additional information (n = 203)	Assigned: Logicalscientific (n = 200)	Assigned: Narrative (n = 199)	Self-selection: Logical scientific (n = 159)	Self-selection: Narrative (n = 43)
Choice decisions in choice experiment					
% choosing non-browning	26.0	28.8	36.2	33.4	34.1
% choosing antioxidant-enhanced	22.8	24.3	29.4	27.9	23.6
% choosing gene editing	12.3	15.2	20.9	17.4	17.1
% choosing genetic modification	9.0	10.4	13.0	13.8	10.5
% choosing edible coating	13.4	11.8	13.1	11.5	15.1
% choosing conventional (3rd alt.)	43.2	37.0	37.0	35.7	34.9
% choosing no-purchase (4th alt.)	22.1	25.6	15.9	21.5	22.5

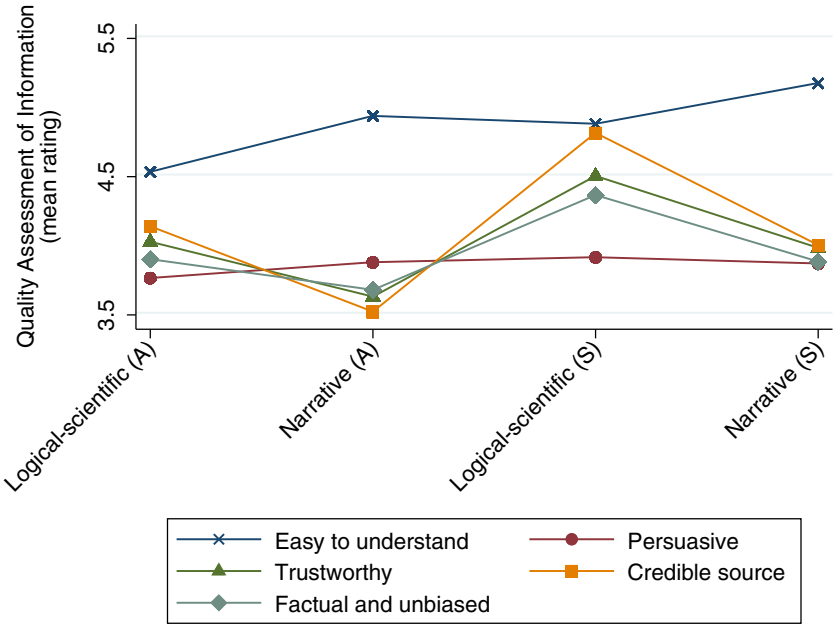

Figure 2. Perceived information quality across treatments. Note: (A) denotes the random assigned conditions, and (S) denotes the self-selected conditions.

as biotechnology information sources (*Government*, *Scientist*). A logistic model reveals the factors that motivate individuals to self-select the logical-scientific rather than a narrative format (table 6).

Individuals' socio-demographic characteristics, pre-existing beliefs about agricultural biotechnology, and their trust in different biotechnology information sources shape the relative appeal of different types of information. Results indicate that, *ceteris paribus*, individuals holding relatively more favorable attitudes toward science in general (*Science*), and who self-reported as more familiar with biotechnology (*Familiar*), are more likely to choose to read one of the logical-scientific information sources in the self-selection condition. However, unlike Huffman *et al.* (2004) who suggested significant impacts of socio-demographics on individuals' preferences for different sources of information on genetic modification, we find that individuals' socio-demographic characteristics and trust in different information sources are not related to their information choices.

Narrative Effects in Choice Behavior

The discrete choice experiment allows us to examine how consumers respond to the enhanced apple characteristics (non-browning and antioxidant enhanced), to the technology used to produce these characteristics (gene editing, genetic modification, edible coating), and most importantly, whether exposure to either a narrative information or a logical-scientific treatment prior to completing the choice experiment produces different responses to the novel food technologies and apple characteristics.

The lower panel of table 5 presents the summary statistics for the choice decisions made in each information treatment. Respondents assigned to the "narrative" condition chose the novel apple attributes and technologies – non-browning, antioxidant-enhanced, and gene editing – more frequently relative to other treatments; these respondents are also less likely to choose the "no-purchase" option in the choice experiment. In addition, in all information treatments, over a third of respondents preferred the "conventional" option – sliced apples turn brown quickly, contain a regular amount of antioxidants, and are produced without using any food technologies (gene editing, genetic modification, edible coating). Respondents assigned to the "no additional information" treatment most frequently chose this status-quo "conventional" alternative, whereas they least frequently choose apples with non-browning, antioxidant-enhanced, gene editing, and genetic modification attributes.

To produce more precise and efficient estimates, we tested whether the data can be pooled

Table 6. Motivating Factors for Self-Selection of Logical-Scientific Versus Narrative Format

Variable[1]	Description	Par.	Std. err.
Intercept		−1.72	1.52
Socio-demographic			
Male	1 if male; 0 otherwise	0.05	0.44
Age	Age in years	0.00	0.02
University	1 if obtained university degree or higher; 0 otherwise	0.24	0.43
Inc	1 if annual household income greater than CAD$125,000; 0 otherwise	−0.21	0.51
Kid	1 if children under 18 years old live in household; 0 otherwise	0.21	0.52
Biotechnology beliefs			
Science	Attitudes to science and technology in general: 1 = the world is lot worse off because of S & T; 6 = the world is lot better off because of S & T	0.32*	0.19
Familiar	Familiarity with biotechnology:1 = know some or a lot about biotechnology; 0 = know nothing at all or just a little	1.00**	0.46
Risk	1 if believe the risks of biotechnology outweigh benefits; 0 otherwise	−0.58	0.44
Trust in biotechnology information sources			
Government	1 = no trust at all; 6 = very high trust	0.27	0.18
Scientist		−0.02	0.25
Consumer organization		0.02	0.19
Science journalist		0.04	0.21
Model fit			
Log likelihood		−83.42	
Pseudo R^2		0.10	
# of par.		12	
N		176[2]	

Note: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

[1] Dependent variable is the format of self-selected information: = 1 if self-selected the logical-scientific information (a government or scientific organization); = 0 if self-selected the narrative information (a consumer blogger or newspaper journalist).

[2] Among the 202 respondents assigned to self-selection condition, twenty-six respondents did not provide complete socio-demographic information.

or combined across treatments. Specifically, we test whether respondents exhibit similar preference structures given different information sources (government vs. a scientific organization, a consumer blogger vs. a newspaper journalist), for a given information format (logical-scientific or narrative), and exposure condition (random allocation or self-selection). Following Swait and Louviere, Fox, and Moore (1993), we use a manual grid search approach to identify the relative scale parameter between source treatments and then use a two-step likelihood ratio (LR) test to test the null hypothesis of preference and scale equality across source treatments. For instance, we tested whether there are differences in preference and scale parameters between respondents who self-selected a consumer blogger or a media journalist source of narrative information. The *LR* test, taking into account of the relative scale factor ($\bar{\lambda} = 0.585$), yields a $\chi^2_{df=6}$ value of 9.63, suggesting that we cannot reject the null hypothesis of equivalent preference parameters. Conditional on the equal preference parameters, a further *LR* test of equal scale parameters yields a $\chi^2_{df=1}$ value of 7.85, indicating that we can reject the null hypothesis of equal scale parameter at a 99% confidence level. Results in table 7 present sufficient evidence that data across respective source treatments exhibit similar preference structures, thus pooling and jointly estimating data across source treatments, with appropriate rescaling of data, are justified. Consequently, the number of information treatments reduces from nine to five (see figure 1).

Tables 8 and 9 report estimation results for both the MNL and RPL models. Both joint models suggest that respondents perceive the

Table 7. Estimation and Testing Results for Pooling Information Source Treatments

Information source treatment	Estimate of relative scale factor ($\hat{\lambda}$)	L_1	L_2	L_λ	*LR* test	*p*-value[1]	Parameter equality	L_p	*LR* test	*p*-value[2]	Scale equality
Assigned condition											
H_0: Pooling over government (1) and scientist (2) is OK	0.920	−778.56	−773.59	−1553.00	1.70	0.945	Fail[3]	−1554.85	3.71	0.054	Fail
H_0: Pooling over consumer (1) and journalist (2) is OK	1.005	−775.39	−736.31	−1513.13	2.86	0.826	Fail	−1513.13	0.01	0.905	Fail
Self-selection condition											
H_0: Pooling over government (1) and scientist (2) is OK	1.055	−563.64	−661.66	−1229.89	9.18	0.164	Fail	−1230.49	1.19	0.275	Fail
H_0: Pooling over consumer (1) and journalist (2) is OK	0.585	−55.84	−274.57	−335.23	9.63	0.141	Fail	−339.15	7.85	0.005	Reject

[1] p-value calculated for 6 d.f. and the null hypothesis tested at 95% confidence level.
[2] p-value calculated for 1 d.f. and the null hypothesis tested at 95% confidence level.
[3] "Fail" denotes that we cannot reject the null hypothesis of equivalent preference or scale parameters.

non-browning attribute for sliced apples positively. The antioxidant-enhanced attribute is only marginally significant (10% level) in the RPL model, suggesting that this is not a particularly compelling benefit for this sampled group of consumers. Respondents discount all three novel food technologies – gene editing, genetic modification, and edible coating – compared to the conventional production method, *ceteris paribus*, and this result holds across both types of models. Gene editing is the least discounted, followed by edible coating, whereas genetic modification is the most discounted food technology.[viii] As expected, a higher price is associated with a lower utility level. Respondents prefer to select from one of the three alternatives provided rather than buying nothing, as shown by the negative and statistically significant parameter estimate for *no-purchase*.

Of primary interest is the impact of information format on consumers' responses to apple characteristics and food technologies. Following Louviere, Hensher, and Swait (2000, p. 364) and Louviere, Fox, and Moore (1993, p. 211), we reject the joint null hypothesis that the preference structures are equivalent across all information treatments ($H_0 : \beta_{NO} = \beta_{LS_A} = \beta_{NR_A} = \beta_{LS_S} = \beta_{NR_S} = \beta$; $\chi^2_{df=24} = 82.85$ and $p < 0.01$ for MNL; $\chi^2_{df=48} = 110.37$ and $p < 0.01$ for RPL). That is, information format matters.

For any given information condition, the *LR* tests indicate that the RPL model allowing for preference heterogeneity outperforms the MNL model. In addition, the statistical significance of standard deviation estimates in the RPL model indicates that respondents are heterogeneous in their preferences for the novel apple characteristics and food technologies. Comparing preference parameters across information treatments is less insightful than comparing the WTP estimates, which are not confounded with the scale parameters. Hence, we examine WTP values as another means of evaluating the differences across information treatments. We derive WTP estimates from the RPL model results (table 10).

We test differences in WTP values across information treatments with the

[viii] Both models suggest that gene editing is significantly preferred over genetic modification at the 1% significance level (Wald test). The preference for gene editing over edible coating is also significant at 1%. Edible coating is preferred to genetic modification but only at a 10% significance level.

Table 8. MNL Model Estimates across Information Treatments

			Assigned		Self-selection	
Attribute[1]	Joint	Control (NO)	Logical scientific[2] (LS_A)	Narrative[2] (NR_A)	Logical scientific[2] (LS_S)	Narrative[2] (NR_S)
Non-browning	0.69*** (0.07)[3]	0.56*** (0.14)	0.69*** (0.13)	0.65*** (0.12)	0.84*** (0.14)	1.20*** (0.45)
Antioxidant-enhanced	0.18*** (0.06)	0.19 (0.13)	0.26** (0.12)	0.02 (0.12)	0.31** (0.13)	−0.39 (0.38)
Gene editing	−1.04*** (0.09)	−1.40*** (0.18)	−1.11*** (0.18)	−0.58*** (0.16)	−1.07*** (0.19)	−1.04* (0.58)
Genetic modification	−1.53*** (0.09)	−1.80*** (0.19)	−1.60*** (0.19)	−1.18*** (0.18)	−1.37*** (0.20)	−2.01*** (0.61)
Edible coating	−1.37*** (0.09)	−1.27*** (0.18)	−1.43*** (0.18)	−1.19*** (0.17)	−1.63*** (0.20)	−1.30** (0.58)
Price	−0.66*** (0.04)	−0.66*** (0.07)	−0.60*** (0.07)	−0.69*** (0.07)	−0.69*** (0.08)	−0.46*** (0.15)
No-purchase	−3.32*** (0.15)	−3.47*** (0.32)	−2.92*** (0.32)	−3.79*** (0.30)	−3.42*** (0.35)	−1.77*** (0.47)
Relative scale factor ($\bar{\lambda}$)	–	1.000	1.040	0.870	0.970	0.955
Log-likelihood	−6204.99	−1529.87	−1554.85	−1513.13	−1230.49	−335.23
# of respondents	804	203	200	199	159	43
# of choices[4]	4824	1218	1200	1194	954	258

Note: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.
[1] All attribute variables are dummy coded.
[2] Estimates are adjusted for scale differences.
[3] Numbers in parentheses are standard errors.
[4] Each of the 804 respondents made six choice decisions.

Table 9. RPL Model Estimates across Information Treatments

			Assigned		Self-selection	
Attribute[1]	Joint[2]	Control (NO)	Logical scientific (LS_A)	Narrative (NR_A)	Logical scientific (LS_S)	Narrative (NR_S)
Non-browning	0.41** (0.16)[3]	0.33 (0.31)	−0.06 (0.41)	0.75** (0.30)	0.68* (0.36)	0.55 (0.70)
Antioxidant-enhanced	−0.25* (0.14)	−0.31 (0.30)	0.01 (0.31)	−0.35 (0.26)	−0.05 (0.37)	−0.95 (0.61)
Gene editing	−1.46*** (0.16)	−2.18*** (0.35)	−1.87*** (0.38)	−0.69** (0.27)	−1.94*** (0.40)	−0.73 (0.72)
Genetic modification	−2.39*** (0.18)	−2.80*** (0.36)	−2.80*** (0.43)	−1.96*** (0.34)	−2.24*** (0.40)	−2.17** (0.85)
Edible coating	−2.10*** (0.17)	−1.68*** (0.29)	−2.25*** (0.36)	−2.10*** (0.37)	−2.98*** (0.48)	−1.94* (1.06)
Price	−1.05*** (0.05)	−0.96*** (0.10)	−0.99*** (0.11)	−1.20*** (0.11)	−1.23*** (0.17)	−0.55** (0.23)
No-purchase	−5.09*** (0.23)	−4.80*** (0.44)	−4.64*** (0.47)	−5.98*** (0.47)	−5.77*** (0.55)	−2.82*** (0.98)
Standard Deviation						
σ_{NB}	3.15*** (0.18)	2.57*** (0.28)	3.85*** (0.41)	2.75*** (0.29)	3.38*** (0.45)	3.04*** (0.75)
σ_{AE}	2.39*** (0.14)	2.38*** (0.27)	2.34*** (0.32)	2.27*** (0.26)	2.55*** (0.35)	2.03*** (0.54)
σ_{GE}	1.42*** (0.16)	1.61*** (0.38)	1.96*** (0.36)	0.76* (0.40)	1.92*** (0.49)	1.20 (0.74)
σ_{GM}	1.36*** (0.18)	1.02** (0.44)	1.81*** (0.44)	1.50*** (0.33)	1.22*** (0.40)	1.28 (1.00)
σ_{EC}	1.54*** (0.17)	0.80** (0.39)	1.22*** (0.34)	1.99*** (0.36)	1.73*** (0.53)	3.67*** (1.24)
Log-likelihood	−5181.54	−1303.95	−1258.77	−1276.10	−1007.52	−280.02
# of respondents	804	203	200	199	159	43
# of choices[4]	4824	1218	1200	1194	954	258

Note: *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.
[1] All attribute variables are dummy coded.
[2] We estimate the RPL model by the maximum simulated likelihood with 200 Halton draws.
[3] Numbers in parentheses are standard errors.
[4] Each of the 804 respondents made six choice decisions.

Table 10. Comparison of WTP Estimates across Information Treatments: RPL Model

	Non-browning		Antioxidant-enhanced		Gene editing		Genetic modification		Edible coating	
No vs. A_Logic[1]	0.34[2]	−0.06	−0.33	0.01	−2.26***	−1.87***	−2.90***	−2.81***	−1.74***	−2.26***
No vs. S_Logic	0.34	0.56*	−0.33	−0.04	**−2.26***[3]	**−1.57***	**−2.90***	**−1.81***	−1.74***	−2.41***
No vs. A_Narr	0.34	0.62**	−0.33	−0.29	**−2.26***	**−0.57**	**−2.90***	**−1.63***	−1.74***	−1.75***
No vs. S_Narr	0.34	0.99	−0.33	−1.71	−2.26***	−1.31	−2.90***	−3.93	−1.74***	−3.51
A_Logic vs. S_Logic	−0.06	0.56*	0.01	−0.04	−1.87***	−1.57***	**−2.81***	**−1.81***	−2.26***	−2.41***
A_Logic vs. A_Narr	−0.06	0.62**	0.01	−0.29	**−1.87***	**−0.57**	**−2.81***	**−1.63***	−2.26***	−1.75***
A_Logic vs. S_Narr	−0.06	0.99	0.01	−1.71	−1.87***	−1.31	−2.81***	−3.93	−2.26***	−3.51
S_Logic vs. A_Narr	0.56*	0.62**	−0.04	−0.29	**−1.57***	**−0.57**	−1.81***	−1.63***	**−2.41***	**−1.75***
S_Logic vs. S_Narr	0.56*	0.99	−0.04	−1.71	−1.57***	−1.31	−1.81***	−3.93	−2.41***	−3.51
A_Narr vs. S_Narr	0.62**	0.99	−0.29	−1.71	−0.57**	−1.31	−1.63***	−3.93	−1.75***	−3.51

Note: *, **, and *** denote the WTP values are statistically different from zero at the 10%, 5%, and 1% significance levels, respectively.

[1] A = assigned to the information condition, S = self-selected the information condition.

[2] WTP values are derived from the RPL model in table 9, for 500 g bag of apple slices. Reported statistics are point estimates obtained from 1,000 bootstrapped WTP estimates using the Krinsky–Robb (1986) bootstrapping method.

[3] We test differences in WTP values across information treatments with the nonparametric combinatorial method suggested by Poe, Giraud, and Loomis (2005). Bold values denote the WTP differences are statistically significant at the $p < 0.10$ level.

nonparametric combinatorial method suggested by Poe, Giraud, and Loomis (2005). Results lend reasonably strong support for the existence of information framing effects for the two biotechnologies: gene editing and genetic modification. However, information format appears to have little impact on WTP values for the non-browning, antioxidant-enhanced, and edible coating attributes. This is unsurprising as the primary focus of the provided information was to explain to respondents how gene editing works differently from genetic modification to achieve the novel apple characteristics, although edible coating was not discussed in detail.

Relative to being allocated to a narrative information treatment (*A-Narr*) and voluntarily selecting to the logical-scientific information (*S-Logic*), respondents who did not receive any additional information (*No*) tended to discount the gene editing technology more heavily, as shown by the significantly smaller WTP estimates (-$2.26 for *No*, -$1.57 for *S_Logic*, and -$0.57 for *A_Narr*). This was also the case for genetic modification (-$2.90 for *No*, -$1.81 for *S_Logic*, and -$1.63 for *A_Narr*). Interestingly, respondents randomly allocated to a logical-scientific information treatment (*A_Logic*) did not behave differently from those without any additional information (*No*). The failure of respondents to assimilate the provided logical-scientific information and to adjust their decision making accordingly may reflect inattention or lack of motivation to process this type of information (see Golman, Hagmann, and Loewenstein [2017] for a review on various incentives for information avoidance).

Turning to a comparison of the narrative information format, we find that, relative to respondents who were allocated to a logical-scientific information treatment (*A_Logic*), those allocated to view the narratives (*A_Narr*) viewed both gene editing and genetic modification more positively, as reflected in the significantly larger (i.e. less negative) WTP estimates (-$0.57 vs. -$1.87 for gene editing, -$1.63 vs. -$2.81 for genetic modification). Such a narrative effect is also significant relative to the condition where respondents actively chose to view the logical-scientific information (*S_Logic*, -$0.57 vs. -$1.57).

The self-selection condition allows us to compare the effect of information when the respondent chooses the type of information to view rather than a random assignment to

read either logical-scientific or narrative information. Although there appears to be no difference in WTP values between respondents randomly allocated to logical-scientific information (*A_Logic*) and actively choosing this type of information (*S_Logic*) in the case of a new and unfamiliar technology such as gene editing, the significantly larger WTP estimate (-$1.81 vs. -$2.81) suggests that respondents are more likely to be receptive to information about genetic modification framed as a logical-scientific discourse from governmental agencies or scientific bodies when they actively seek out this type of information. However, we identify no such self-selection effects for narratives.

In sum, the analysis suggests that *how* information about novel agricultural and food technologies is communicated to consumers matters. As expected, although additional positive information can reduce resistance to a new technology, presenting that information in the style of a narrative (relative to a logical-scientific framing) appears to be particularly effective. In addition, the way in which this information is accessed matters (assigned versus actively choosing a type and source of information).

Conclusion

Although previous economics literature has identified information framing effects in a food context, this article is the first to demonstrate the power of narratives in biotechnology communication. The advent of gene editing (CRISPR-Cas9) technology and its potential application across a wide range of plant and animal breeding contexts provides a timely opportunity to explore how the framing of information about this technology as a narrative influences consumer and public acceptance.

Unsurprisingly, analysis of choice behavior reveals that all three novel food technologies – two plant breeding techniques gene editing and genetic modification, and one food processing method edible coating – are discounted by consumers, a result consistent with prior research on technology neophobia. What is more interesting, and potentially of some comfort to agricultural and food scientists, is that the resistance to gene editing technology appears to be significantly milder than to genetic modification and edible coating, suggesting a "window of acceptance" may be available in which to provide constructive, positive communication about this new technology.

In terms of how to frame communication about new technologies such as gene editing, the results show that format and framing of information matter a great deal. Factually equivalent information about food technology framed in different formats (logical scientific vs. narrative) or accessed by respondents through different pathways (assigned or voluntary choice) has different influences on consumer choices. Compared with logical-scientific information, narratives help to ameliorate negative responses to food biotechnology.

Our results have broader implications for policymakers, scientists, and industry stakeholders from a public communication perspective. Narratives developed from factual and unbiased scientific knowledge may assist with public communication as they promote a greater comprehension and engagement within the science debate; they could also be used to ameliorate the negative consequences of misinformation spread through mass or social media where the use of narratives is common. Mass media and interactive social media platforms have furnished consumers with a flood of information about biotechnology, which at times contradicts scientific consensus. Scientists, as trustworthy communicators of biotechnology information, could potentially increase their visibility in the biotechnology dialogue by creating credible narrative messages.

Given the potential for substantial narrative effects, policymakers, and industry stakeholders may encounter ethical considerations in using narratives to communicate science, as their impacts are difficult to counteract (Dahlstrom and Ho 2012). From a public policy perspective, the objective of this communication should be to enhance public understanding rather than to impose a predefined opinion (Weitze and Puhler 2013). The empirical analysis also suggests that narratives attributed to ostensibly less credible sources (i.e. a consumer or journalist) could have stronger impacts in shaping attitudes and behaviors, compared with logical-scientific information attributed to more credible sources (i.e. scientists or government). As such, the responsible use of narratives will be important in preventing the provision of misleading information to consumers.

The narrative message in this article features a testimonial from a mother talking about her experience with novel apple traits and food technologies (Waltz 2015). The testimonial narrative presents a one-sided

perspective by highlighting only the benefits but ignoring any risks or uncertainty aspects of biotechnology. It is possible that the manipulative nature of this type of narrative message may discount the potential narrative effects (Slater *et al.* 2003). Future studies could instead adopt a conversational narrative, presenting different perspectives from the biotechnology debate by including multiple story narrators both endorsing and opposing biotechnology. Additionally, despite the availability of diverse narrative formats, this analysis focuses on a written text-based format for biotechnology communication. Further research could explore the impacts of visual narratives, such as documentaries, YouTube videos, and interactive video games related to biotechnology, which are assumed to have a greater narrative power (Downs 2014).

We deliberately constructed the combination of information source and format (i.e. logical-scientific information attributed to a government agency or scientific body, narratives attributed to a consumer blogger or newspaper journalist) to conform to normative expectations (e.g. scientists are expected to use logical-scientific language in public communication).[ix] An interesting angle for further study would be to violate this normative expectation (e.g. letting scientists tell stories) and thus examine consumers' responses to unexpected source-and-format combinations, as well as impacts on source trustworthiness and message credibility.

Finally, this article demonstrates the power of using narratives in biotechnology communication where some public opinions have already been established (i.e. in the case of genetic modification), as well as a relatively new technology (gene editing). The narrative communication approach for other technologies that are at an early point in development, such as food nanotechnology, warrants further research.

[ix]As pointed out by an anonymous reviewer, the manipulated combination of information source and format contributes to the credible experimental design, however, it is also a potential drawback of the design as the identified narrative effect may confound with the effect of information sources. Indeed, respondents perceived the narrative messages attributed to a consumer blogger or newspaper journalist as "*from less credible sources*" than the logical-scientific messages from government agency or scientists (figure 2). Although the current design does not allow us to identify each effect separately, the power of the narrative effect is still justified as we show that the narratives from less credible sources have significant impacts on consumer attitudes and choice behaviors. Thus, the narrative effect may dominate the potential source effect.

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Appendix A

Information Conditions

General information received by all survey respondents:

Apples have been symbolized as a healthy food choice. Health Canada recommends 7 to 10 servings of fruits and vegetables a day depending on age and gender. However, only 26% of Canadians meet this daily intake recommendation.

Among the reasons often given for not consuming apples are price, inconveniently

large size which makes eating a whole apple messy, apples turn brown so quickly when sliced, and the attractiveness of other fruits with preferred flavor or nutrition.

Scientists have been making efforts – through biotechnology – in developing apple varieties with additional benefits: a non-browning apple, and an apple with enhanced levels of antioxidants, such as Vitamin C.

Information Condition 1: No Additional Information (Control)

Information Condition 2: Logical-scientific

(attributed either to Health Canada or the Academy of Science, The Royal Society of Canada)

Government_Instruction: Suppose the following information about new apple varieties is available from Health Canada. You may take your time in reading this material carefully. When you are done, please proceed to answer some questions about these apple varieties.

OR

Scientist_Instruction: Suppose the following information about new apple varieties is available from the Academy of Science | The Royal Society of Canada. You may take your time in reading this material carefully. When you are done, please proceed to answer some questions about these apple varieties.

Next Generation Biotechnology for Apple Improvements

Health Canada

OR

Biotechnology Solutions for Apple Challenges

Academy of Science | The Royal Society of Canada

Agricultural biotechnology is a collection of scientific techniques used to improve crops and foodstuffs. Scientists have been using biotech tools to develop new apple varieties with desired characteristics, such as non-browning apples and apples enriched with antioxidants.

Apples that resist browning are valuable because they keep their color longer when sliced, reduce waste due to cosmetic "defects," encourage healthy snacking on fresh apple slices, and need no additional anti-browning treatments (e.g. edible coating using a calcium ascorbate solution). In addition, apples are a rich source of dietary antioxidants such as Vitamin C, which helps to protect against the damage of free radicals that cause aging and disease. Enhancing the levels of antioxidants can boost the health benefits of apples.

Conventional plant breeding methods such as crossbreeding are limited to random rearrangement of existing genes between the same or closely related plant species and can take years to achieve the desired characteristics. Biotechnology enables plant improvements that are not possible with conventional breeding, and offers a range of plant breeding tools that are more targeted, precise, and rapid.

The best known biotech tool is genetic modification, which requires the insertion of foreign genes from a different species into the plant to obtain the desired characteristics. As a new biotech tool, the gene editing technique works differently than genetic modification: it locates a gene to be edited then makes the necessary changes, either a deletion or a repair.

To create new apple varieties, the gene editing technique merely changes a few elements of an apple's existing genes. By locating and suppressing the genes responsible for browning, gene-edited apples produce substantially fewer browning-causing enzymes, thus they do not turn brown quickly when sliced. Similarly, by enhancing the activities of genes responsible for antioxidants production, gene-edited apples can contain higher levels of various dietary antioxidants such as Vitamin C.

◦ I have carefully read the information provided and would like to proceed with the survey.

Information Condition 3: Narrative

(attributed to either a consumer blog post or a science journalist with *The Globe and Mail*)

Consumer Blog Post_Instruction: Suppose the following information about new apple varieties is available from Alison Harris, who is a consumer reporting her experience with the new apple varieties in a blog post. You may take your time in reading her story carefully. When you are done, please proceed to answer some questions about these two apple varieties.

OR

Media_Instruction: Suppose the following information about new apple varieties is available from Emma Cooper, who is a science journalist writing in *The Globe and Mail*. You may take your time in reading the newspaper article carefully. When you are done, please proceed to answer some questions about these apple varieties.

The Day My Family Tried Out a New Biotech Apple

Alison Harris, consumer, Blog Post

OR

Tasting and Testing a New Apple Variety – A Home Biotech Test

Emma Cooper, science journalist, The Globe and Mail

Innovations in biotech crops aren't known for getting families excited, but there are new fruit traits that families may actually notice and appreciate. They are apples that do not turn brown when sliced and are enriched with healthy antioxidants. My children and I had a chance to try some last week and see how child friendly and tasty they really are — a snack and a science lesson in one.

My children inhale fresh apples, but when I put them in a lunch box it's another story. Browned from slicing or bruising, by lunch time, apples do not look appetizing and inevitably return home uneaten. The biotech non-browning apples would solve the problem and will not require the addition of browning inhibitors to keep them visually appealing. My family and I were keen to give it a try.

We sliced them, ran them through the blender, and gave them the ultimate backpack test. With each experiment we compared them in taste and visual appeal with conventional apples. The result: biotech apples withstood everything we threw at them, whereas the conventional apples looked, well, brown.

The difference was most obvious when we made smoothies. The conventional apples turned brown almost immediately in the blender, but the biotech apples stayed a bright yellowish-green. Then came the backpack test. I put both a biotech and a conventional variety into a satchel and let my son swing it around on the porch, banging into everything. Then we sliced up both apples. Both were beat up, but the conventional apple had brown bruises at each point of contact, but the biotech variety did not change color!

Scientists accomplished this with a gene editing technology, which locates the gene to be edited and then makes necessary changes, resembling the find-and-replace function on the computer. To create the non-browning apple, the gene editing technique locates the gene responsible for apple browning and then snips out (thus disables) a tiny piece of the gene.

Unlike genetic modification, gene editing does not require inserting foreign genes from other species into the apple and merely works on an apple's existing genes. Using the same technique, researchers are also working on creating apples with improved health benefits such as apples enriched with healthy antioxidants like Vitamin C.

◦ I have carefully read the information provided and would like to proceed with the survey.

Information Condition 4: Self-Selection

Now, suppose the following information about new apple varieties becomes available to you. The titles and sources for each of these pieces of information are presented below. Please choose one of these information sources and take your time in reading the information carefully (remember you may choose only one from the four options below). When you are done, please proceed to answer some questions about these two apple varieties.

◦ **Next Generation Biotechnology for Apple Improvements**
Health Canada

◦ **Biotechnology Solutions for Apple Challenges**
Academy of Science | The Royal Society of Canada

◦ **The Day My Family Tried Out a New Biotech Apple**
Alison Harris, consumer, Blog Post

◦ **Tasting and Testing a New Apple Variety – A Home Biotech Test**
Emma Cooper, science journalist, The Globe and Mail

Appendix B

Perceptions of Information Treatments

Table B1 Evaluations of Information Quality across Treatments

	Assigned				Self-selection			
	Logical scientific		Narrative		Logical scientific		Narrative	
	Government (n = 100)	Scientist (n = 100)	Consumer (n = 102)	Journalist (n = 97)	Government (n = 73)	Scientist (n = 86)	Consumer (n = 8)	Journalist (n = 35)
Easy to understand	4.45	4.59	4.89	4.96	**4.55**	**5.14**	5.38	5.11
Persuasive	3.65	3.86	3.75	3.99	3.75	4.03	2.88	4.09
Trustworthy	3.92	4.12	3.42	3.84	4.36	4.62	3.25	4.14
From a credible source	3.92	4.35	**3.13**	**3.93**	4.70	4.91	3.63	4.09
Factual and unbiased	**3.65**	**4.15**	**3.45**	**3.92**	4.26	4.45	3.63	3.94

Note: significant differences (1% significant level) between treatment sources – government vs. scientific organization, consumer blogger vs. journalist – are shown in bold.